

Laboratory work 3 report
Khamrayeva Rabina
CSS4007-ENG-6

What RAG Is Made Of

Purpose of work

In this laboratory work, I studied the main components of the RAG system: chunking, embedding, retrieval and generation. I used a small case of four fictional articles of banking policy and checked how well the system finds the information and uses it to generate an answer.

1. Chunking

First I divided the case in two ways: at the level of whole articles and at the level of individual proposals. There were four articles in the case containing a total of 22 proposals. So I expected to get 4 chunks at article-level chunking and 22 chunks at sence-level chunking.

The actual result coincided with the expectation: 4 article-level chunks and 22 sence-level chunks were received.

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article-level chunks: 4  
sence-level chunks: 22
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2. Embedding and Retrieval

For retrieval I used the intfloat/multilingual-e5-small model that converts text into vectors. Then, to compare the query with chunks used cosine similarity.

For the first request: "How much does an instant replacement card at the branch?" I expected article-level retrieval will find the article cards-loss, and sence-level retreat will find a proposal for a moment card worth 3000 tenge. Article-level retrieval really returned cards-loss from score 0.869. However, at the level of proposals, the first proposal was to replace the card within 5 working days from score 0.851. The correct offer on the instant map for 3000 tenge was the second with score 0.848.

```
query: How much does an instant replacement card cost at the branch?  
article-level retrieved: cards-loss (score 0.869)  
sentence-level [0.851] cards-loss:4: A replacement card is issued free of charge and is ready at your chosen branch within 5 working days.  
sentence-level [0.848] cards-loss:5: If you need a card the same day, the branch can issue an instant card for 3000 tenge.  
sentence-level [0.828] cards-expiry:4: Uncollected cards are destroyed after 60 days for security reasons, and a new one then costs 1500 tenge.
```

For paraphrase-query: "I lost my wallet this morning and need a working card by tone - what will it me?" I also expected that the card-loss article will be found. However, article-level retrieval returned cards-Empire with score 0.881. At sence-level the first three results were cards-expiry:4 with score 0.864,

cards-loss:4 with score 0.861 and cards-pin:1 with score 0.860. The correct offer about 3000 tenge in Top-3 has not hit.

This showed that paraphrase-the request was more difficult for retrieval-model than direct literal-request.

```
query: I mistyped my code at the till and the card stopped working - once I sort it out on my phone, how soon can I pay with it again?
article-level retrieved: cards-lost (score 0.875)
sentence-level [0.865] cards-lost:0: If your card is lost or stolen, block it at once in the mobile app under Cards, then Block, or call the round-the-clock hotline printed on the back of the card.
sentence-level [0.864] cards-contactless:2: You can lower the limit, or switch contactless off completely, in the app under Card settings.
sentence-level [0.853] cards-expiry:2: The new card keeps the same number and PIN, but has a new expiry date and a new security code.
```

3. Generation

In the third part I checked the response generation without retrieved context and with retrieved context.

Without context, the model did not call the correct cost and replied that it could not provide information about a particular service and suggested to contact the official website or support.

With retrieved context model gave the correct answer: "The boss of an instant replacement card at the branch is 3000 tenge."

Thus, without context, the model had no information about the rules of fictional bank, and after the transfer of the desired fact was able to use it for the correct answer.

```
--- WITHOUT retrieved context ---
It typically takes about 24 hours for the card to fully function again after you unblock it through the app.

--- WITH retrieved context ---
It takes effect within 2 minutes.
```

4. Additional check Retrieval

In detailed task I checked the card unlock request: "How long does it take for the card to work again after I unblock it in the app?"

Article-level retrieval correctly defined article cards-pin-block with score 0.908. However, sentence-level retrieval did not find the right offer in Top-3. The first was the offer to block the map from score 0.887, although the issue was about unlocking. This shows that retrieval can confuse similar according to words, but opposite in the meaning of action.

Also I calculated Recall@1 for all 8 requests. Article-level retrieval correctly found the right article in 6 of 8 cases, so Recall@1 was 0.75. Sentence-level retrieval correctly found the right offer in 3 of 8 cases, so Recall@1 was 0.375. For literal query about the snap map I also checked the proximity of the first results. The wrong first sentence had score 0.8514, and the correct 0.8476. The difference was only 0.0038. This shows that cosine similarity is a similarity indicator and is used to rank results, but is not the probability of the answer correctly.

Core Task 1

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query: How long does it take for the card to work again after I unlock it in the app?
article-level retrieved: cards-pin-block (score 0.908)
sentence-level [0.887] cards-lost:1: Blocking takes effect within 60 seconds, after which the card cannot be used for payments or cash withdrawals.
sentence-level [0.879] cards-lost:8: If your card is lost or stolen, block it at once in the mobile app under Cards, then Block, or call the round-the-clock hotline printed on the back of the card
sentence-level [0.878] cards-pin-block:2: To unlock it, open the app, choose the card, and confirm with the six-digit code we send by SMS.
```

Core Task 2

```
Article-level Recall@1: 6 / 8 = 0.75
Sentence-level Recall@1: 3 / 8 = 0.375
```

Core Task 3

```
0.8514 cards-lost:4: A replacement card is issued free of charge and is ready at your chosen branch within 5 working days.
0.8476 cards-lost:5: If you need a card the same day, the branch can issue an instant card for 3000 tenge.
0.8277 cards-expiry:4: Uncollected cards are destroyed after 60 days for security reasons, and a new one then costs 1500 tenge.
```

5. Conclusion

In this laboratory work, I have tested the main stages of simple RAG system: breaking documents on chunks, creation of embeddings, retrieval and response generation. I saw that article-level retrieval in my experiment gave Recall@1 0.75, and sence-level 0.375. At the same time, the smaller breakdown did not always lead to the better result: the model sometimes chose a proposal similar to the words, but not the corresponding meaning of the query. It was especially noticeable for blocking and unlocking requests. Also I checked the impact of the context on the generation and saw that when transferring the correct retrieved fact model could give an accurate answer.